Chapter 4. The Binomial Theorem and Related Identities

No Matter How You Slice It

Contents

4.1 The binomial theorem **2**

4.2 The multinomial theorem **4**

4.3 When the exponent is not a positive integer **5**

4.4 Selected exercises **5**

Binomial and multinomial coefficients; most proofs count the same objects in two ways (**double counting**).

4.1 The binomial theorem

Motivation. Expanding $(x+y)^n$ chooses x or y from each factor, so its coefficients count subsets.

Theorem 4.1 (Binomial theorem)

For every nonnegative integer n,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}. \tag{4.1}$$

Proof. Expanding chooses one summand from each of the n factors; the product is $x^k y^{n-k}$ iff x comes from a k-set of factors, in $\binom{n}{k}$ ways. ■

Theorem 4.2

For every positive integer n: $\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0$.

Proof. Put $x = -1$, $y = 1$ in (4.1): the left side is $0^n = 0$. ■

Theorem 4.3 (Pascal's rule)

For all nonnegative integers n, k:

$$\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}. \tag{4.2}$$

Proof. The right side counts $(k+1)$-subsets S of $[n+1]$. If $n+1 \in S$, the rest of S is a k-subset of $[n]$ (first term); if not, S is a $(k+1)$-subset of $[n]$ (second term). ■

Theorem 4.4

For every nonnegative integer n: $2^n = \sum_{k=0}^{n} \binom{n}{k}$.

Proof. Both sides count subsets of $[n]$ (Theorem 2.4; the right grouped by size). Or put $x = y = 1$ in (4.1). ■

The **Pascal triangle** has $\binom{n}{i}$ as i-th entry of row n (from 0).

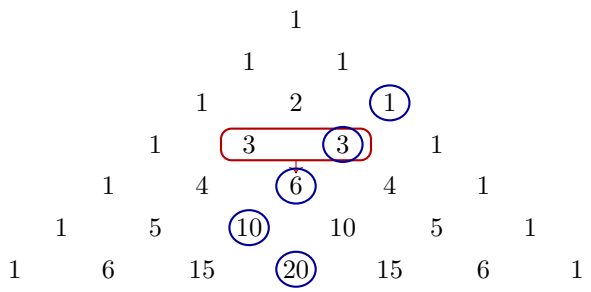


Figure. *Rows 0–6. Red:* $3+3=6$ *(Theorem 4.3). Blue: hockey stick* $1+3+6+10=20$ *(Theorem 4.5).*

Theorem 4.5 (Hockey stick identity)

For all nonnegative integers $k \le n$:

$$\binom{k}{k} + \binom{k+1}{k} + \binom{k+2}{k} + \cdots + \binom{n}{k} = \binom{n+1}{k+1}. \tag{4.3}$$

Proof. Sort the $(k+1)$-subsets of $[n+1]$ by their largest element $k+i+1$ $(0 \le i \le n-k)$: the rest is one of $\binom{k+i}{k}$ k-subsets of $[k+i]$. ■

Theorem 4.6

For every nonnegative integer n:

$$\sum_{k=1}^{n} k\binom{n}{k} = n2^{n-1}. \tag{4.4}$$

Proof.

Step 1 (double counting). Count committees with a president: committee first gives $\sum_k k\binom{n}{k}$; president first, then any subset of the other $n-1$, gives $n2^{n-1}$.

Step 2 (calculus, alternative). Differentiate $(x+1)^n = \sum_{k=0}^{n}\binom{n}{k}x^k$: $n(x+1)^{n-1} = \sum_{k\ge 1} k\binom{n}{k}x^{k-1}$; put $x = 1$. ■

Theorem 4.7 (Vandermonde's identity)

For all positive integers n, m, k:

$$\binom{n+m}{k} = \sum_{i=0}^{k}\binom{n}{i}\binom{m}{k-i}.$$

Proof. Classify k-subsets of $[n+m]$ by the number i of elements in $[n]$: $\binom{n}{i}\binom{m}{k-i}$ each. ■

Theorem 4.8

If $0 \le k \le \frac{n-1}{2}$, then

$$\binom{n}{k} \le \binom{n}{k+1}, \tag{4.5}$$

with equality iff $n = 2k+1$.

Proof. Multiplying $\frac{n!}{k!(n-k)!} \le \frac{n!}{(k+1)!(n-k-1)!}$ by $\frac{k!(n-k-1)!}{n!}$ gives the equivalent $\frac{1}{n-k} \le \frac{1}{k+1}$, i.e. $2k+1 \le n$; equality iff $n-k = k+1$. ■

Corollary 4.9

If $k \ge \frac{n-1}{2}$ (and $k+1 \le n$), then

$$\binom{n}{k} \ge \binom{n}{k+1}, \tag{4.6}$$

with equality iff $n = 2k+1$.

Proof. With $k' := n-k-1 \le \frac{n-1}{2}$ and Proposition 3.18(1), the claim is $\binom{n}{k'} \le \binom{n}{k'+1}$ (Theorem 4.8), with equality iff $n = 2k'+1$, i.e. $n = 2k+1$. ■

$a_0, \dots, a_n$ is **unimodal** if $a_0 \le \cdots \le a_m \ge \cdots \ge a_n$ for some m; so every row of the Pascal triangle is unimodal, peaking in the middle.

4.2 The multinomial theorem

Motivation. Coefficients of $(x_1 + \cdots + x_k)^n$ count multiset permutations.

Example 4.10

$$(x+y+z)^3 = x^3 + y^3 + z^3 + 3x^2y + 3x^2z + 3y^2x + 3y^2z + 3z^2x + 3z^2y + 6xyz. \tag{4.7}$$

Solution. Of the 27 choices, x^2y arises in 3 ways, x^3 in one, xyz in $3 \cdot 2 = 6$.

Definition 4.11 (Multinomial coefficient)

For nonnegative integers with $n = \sum_{i=1}^{k} a_i$,

$$\binom{n}{a_1, a_2, \dots, a_k} := \frac{n!}{a_1!\, a_2! \cdots a_k!}. \tag{4.8}$$

For $k = 2$, $\binom{n}{a_1, a_2} = \binom{n}{a_1}$.

Theorem 4.12 (Multinomial theorem)

For all nonnegative integers n, k,

$$(x_1 + \cdots + x_k)^n = \sum_{a_1 + \cdots + a_k = n} \binom{n}{a_1, \dots, a_k} x_1^{a_1} \cdots x_k^{a_k}, \tag{4.9}$$

the sum over all k-tuples of nonnegative integers with sum n.

Proof. $x_1^{a_1} \cdots x_k^{a_k}$ arises from choosing x_i in a_i factors, i.e. from a word of length n with a_i letters x_i (j-th letter = choice in factor j); there are $\binom{n}{a_1, \dots, a_k}$ (Theorem 3.5). ■

Theorem 4.13

For nonnegative integers with $n = \sum_i a_i$:

$$\binom{n}{a_1, \dots, a_k} = \binom{n}{a_1}\binom{n-a_1}{a_2} \cdots \binom{n-a_1-\cdots-a_{k-1}}{a_k} \tag{4.10}$$

(the last factor is $\binom{a_k}{a_k} = 1$).

Proof. Choose the positions of x_1 ($\binom{n}{a_1}$ ways), then of x_2 among the remaining $n - a_1$, and so on. ■

4.3 When the exponent is not a positive integer

Motivation. Real exponents give Newton's binomial series, needed for generating functions (Chapter 8).

Definition 4.14 (General binomial coefficients)

For real m and integer $k \geq 0$: $\binom{m}{0} := 1$ and $\binom{m}{k} := \frac{m(m-1)\cdots(m-k+1)}{k!}$ for $k > 0$.

Theorem 4.15 (Binomial theorem, general version)

For every real m and $|x| < 1$,

$$(1+x)^m = \sum_{n \geq 0} \binom{m}{n} x^n.$$

Proof [Supplement]. (The book uses Taylor's theorem.)

Step 1 (convergence). For $m \in \mathbb{N}$ use (4.1). Otherwise $\binom{m}{n} \neq 0$ and $\left|\binom{m}{n+1}x^{n+1} / \binom{m}{n}x^n\right| = \frac{|m-n|}{n+1}|x| \to |x| < 1$, so the series converges on $(-1,1)$ to f, $f(0) = 1$, differentiable term by term.

Step 2 (differential equation). Using $(n+1)\binom{m}{n+1} = (m-n)\binom{m}{n}$,

$$(1+x)f'(x) = \sum_{n} \left[(n+1)\binom{m}{n+1} + n\binom{m}{n}\right]x^n = \sum_{n} m\binom{m}{n}x^n = mf(x).$$

Step 3. $\frac{d}{dx}\left[f(x)(1+x)^{-m}\right] = (1+x)^{-m-1}\left[(1+x)f' - mf\right] = 0$, so $f(x)(1+x)^{-m} \equiv 1$. ∎

Example 4.16

$$\frac{1}{\sqrt{1-4x}} = \sum_{n \geq 0} \binom{2n}{n} x^n.$$

Solution. By Theorem 4.15, $(1-4x)^{-1/2} = \sum_n \binom{-1/2}{n}(-4x)^n$ and

$$\binom{-1/2}{n} = \frac{(-\frac{1}{2})(-\frac{3}{2})\cdots(-\frac{2n-1}{2})}{n!} = (-1)^n \frac{(2n-1)!!}{2^n n!}, \tag{4.11}$$

with the **semifactorial** $(2n-1)!! = 1 \cdot 3 \cdots (2n-1)$, $(-1)!! := 1$. So the coefficient is $\frac{2^n(2n-1)!!}{n!} = \frac{(2n-1)!!\,(2\cdot 4\cdots 2n)}{n!\,n!} = \binom{2n}{n}$.

4.4 Selected exercises

Motivation. Book exercises: double counting, consecutive ratios, and reflection of lattice paths.

Exercise 4.4 (Choosing two disjoint subsets)

Let k, m, n be nonnegative integers with $k + m \le n$. Prove that

$$\binom{n}{m}\binom{n-m}{k} = \binom{n}{k}\binom{n-k}{m}.$$

Solution. Count disjoint pairs (M, K) in $[n]$ with $|M| = m$, $|K| = k$.

Step 1 (M first). $\binom{n}{m}\binom{n-m}{k}$.

Step 2 (K first). $\binom{n}{k}\binom{n-k}{m}$.

(Both equal $\binom{n}{m,k,n-m-k}$, Theorem 4.13.) ■

Exercise 4.21 (Log-concavity of binomial coefficients)

(a) Prove that for positive integers $k \le n-1$,

$$\binom{n}{k-1}\binom{n}{k+1} \le \binom{n}{k}^2.$$

(b) (Exercise 4.23.) A sequence $a_0, \ldots, a_n$ of positive reals is *log-concave* if $a_{i-1}a_{i+1} \le a_i^2$ for $1 \le i \le n-1$. Prove that a log-concave sequence is unimodal.

Solution.

(a) For $1 \le k \le n-1$,

$$\frac{\binom{n}{k-1}}{\binom{n}{k}} = \frac{k}{n-k+1}, \qquad \frac{\binom{n}{k+1}}{\binom{n}{k}} = \frac{n-k}{k+1}.$$

so $\binom{n}{k-1}\binom{n}{k+1}/\binom{n}{k}^2 = \frac{k(n-k)}{(k+1)(n-k+1)} < 1$ (strict).

(b)

Step 1 (ratios). $r_i = a_i/a_{i-1}$; dividing $a_{i-1}a_{i+1} \le a_i^2$ by $a_{i-1}a_i > 0$ gives $r_1 \ge r_2 \ge \cdots \ge r_n$.

Step 2 (the peak). Let m be the largest i with $r_i \ge 1$ ($m = 0$ if none). Then $a_{i-1} \le a_i$ for $i \le m$ and $a_{i-1} > a_i$ for $i > m$: $a_0 \le \cdots \le a_m > \cdots > a_n$.

(a) and (b) reprove unimodality of Pascal rows. ■

Exercise 4.24 (Catalan numbers by reflection)

Let C_n be the number of north-eastern lattice paths (steps $(1,0)$ and $(0,1)$) from $(0,0)$ to (n,n) that never go above the diagonal $x = y$. Prove that

$$C_n = \binom{2n}{n} - \binom{2n}{n-1} = \frac{1}{n+1}\binom{2n}{n}.$$

Solution. There are $\binom{2n}{n}$ paths to (n,n). Let $d = y - x$ (each step changes d by ± 1); a path is *bad* if it visits $d = 1$. Paths from $(-1,1)$ to (n,n) number $\binom{2n}{n-1}$; we biject them with bad paths.

Step 1 (reflection). $\varphi(x,y) = (y-1, x+1)$, the reflection in $y = x+1$, sends $d \mapsto 2-d$, swaps east and north steps, and $\varphi(0,0) = (-1,1)$.

Step 2 (bad paths to paths from $(-1,1)$). Reflect a bad path P up to its first point t with $d = 1$: a path from $(-1,1)$ to (n,n).

Step 3 (the inverse). A path Q from $(-1,1)$ goes from $d=2$ to $d=0$, so it visits $d=1$; reflect it up to the first such point. φ preserves $d \neq 1$, so the first point with $d=1$ is unchanged and the maps are inverse.

Step 4 (conclusion). $C_n = \binom{2n}{n} - \binom{2n}{n-1} = \binom{2n}{n}(1 - \frac{n}{n+1}) = \frac{1}{n+1}\binom{2n}{n}$ (ratio from Exercise 4.21(a); $n=0$ directly). ■